

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 7

Groups: The Language of Symmetry

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. Every symmetry group has an identity element. In physical terms, what is the identity symmetry of a crystal? Of a differential equation? Of a deck of cards?

2. Why does the definition of a group require inverses? What kind of symmetry would you be describing if you dropped the inverse axiom?

3. Two groups have the same number of elements but different multiplication tables. Are they “the same” group? What does “the same” mean in group theory?

4. You are told that a physical system has a symmetry group isomorphic to $SO(3)$ (the rotation group in three dimensions). Without knowing anything else, what can you predict about the system's behavior?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
